

basveeling / pcamPublic

<> CodeIssues 8Pull requests 1ActionsProjectsSecurityInsights

master1 branch0 tags

basveeling Merge pull request #10 from Zehui-Lin/master ...

keras_pcamAdded references

.gitignoreinitial

LICENSE.txtinitial

README.mdUpdate README.md

pcam.jpginitial

LocalCodespacesNew

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https://github.com/basveeling/pcam.git

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About

The PatchCamelyon (PCam) deep learning classification benchmark.

#benchmark#deep-learning#dataset#deep-learning-datasets#pathology

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Releases

No releases published

Packages

No packages published

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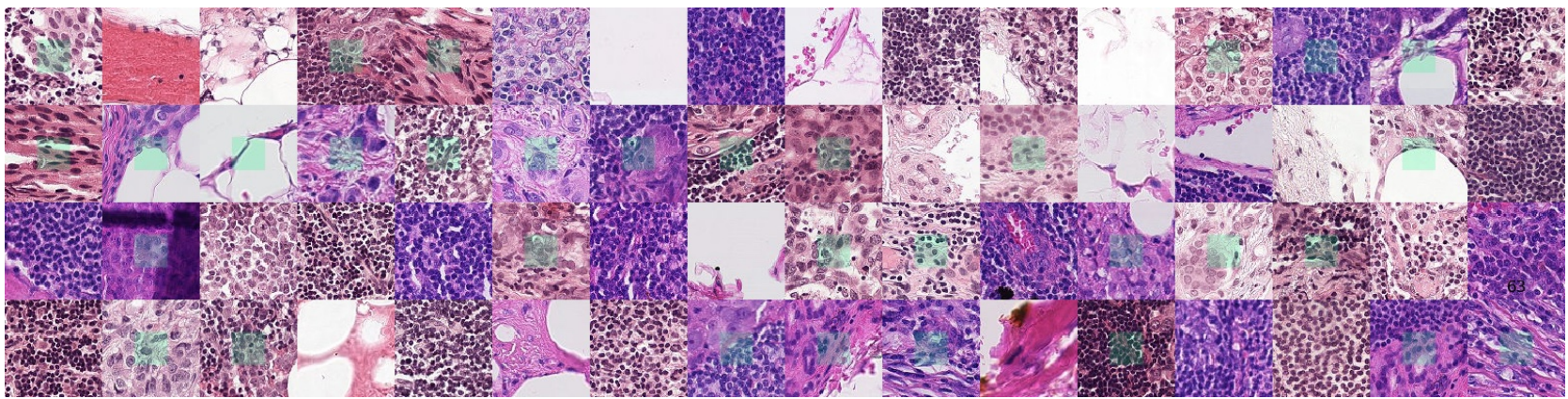
Languages

Python100.0%

PatchCamelyon (PCam)

That which is measured, improves. - Karl Pearson

The PatchCamelyon benchmark is a new and challenging image classification dataset. It consists of 327.680 color images (96 x 96px) extracted from histopathologic scans of lymph node sections. Each image is annotated with a binary label indicating presence of metastatic tissue. PCam provides a new benchmark for machine learning models: bigger than CIFAR10, smaller than imagenet, trainable on a single GPU.



Example images from PCam. Green boxes indicate tumor tissue in center region, which dictates a positive label.

► Table of Contents

Why PCam

Fundamental machine learning advancements are predominantly evaluated on straight-forward natural-image classification datasets. Think MNIST, CIFAR, SVHN. Medical imaging is becoming one of the major applications of ML and we believe it deserves a spot on the list of go-to ML datasets. Both to challenge future work, and to steer developments into directions that are beneficial for this domain.

We think PCam can play a role in this. It packs the clinically-relevant task of metastasis detection into a straight-forward binary image classification task, akin to CIFAR-10 and MNIST. Models can easily be trained on a single GPU in a couple hours, and achieve competitive scores in the Camelyon16 tasks of tumor detection and WSI diagnosis. Furthermore, the balance between task-difficulty and tractability makes it a prime suspect for fundamental machine learning research on topics as active learning, model uncertainty and explainability.

Download

The data is stored in gzipped HDF5 files and can be downloaded using the following links. Each set consist of a data and target file. An additional meta csv file is provided which describes from which Camelyon16 slide the patches were extracted from, but this information is not used in training for or evaluating the benchmark. Please report any downloading problems via a github issue.

Download all at once from [Google Drive](#).

Name	Content	Size	Link	MD5 Checksum
camelyonpatch_level_2_split_train_x.h5.gz	training images	6.1 GB	Download	1571f514728f59376b785fc836ff4d
camelyonpatch_level_2_split_train_y.h5.gz	training labels	21 KB	Download	35c2d7259d986cfc8143347bb8e05f
camelyonpatch_level_2_split_valid_x.h5.gz	valid images	0.8 GB	Download	d8c2d68d498dbd479f8199bdfa8cfd
camelyonpatch_level_2_split_valid_y.h5.gz	valid labels	3.0 KB	Download	68a7835772fbdd7f34eb86d4428cfd
camelyonpatch_level_2_split_test_x.h5.gz	test images	0.8 GB	Download	d5b63478df7cfa627aeec8b9dc8c0b
camelyonpatch_level_2_split_test_y.h5.gz	test labels	3.0 KB	Download	2b85f58b927af9964a4c15b8f7e8f
camelyonpatch_level_2_split_train_meta.csv	training meta		Download	5a3dd671e465cfd74b5b822125e65f
camelyonpatch_level_2_split_valid_meta.csv	valid meta		Download	3455fd69135b66734e1808f3af684f
camelyonpatch_level_2_split_test_meta.csv	test meta		Download	67589e08a4a37ec317f2d1932c75b

Mirror Zenodo:
<https://zenodo.org/record/2546921>

Baidu AI Studio:
<https://aistudio.baidu.com/aistudio/datasetdetail/30060>

Usage and Tips

Keras Example

[General dataloader for keras](#)

README.md

```
from keras.preprocessing.image import ImageDataGenerator

x_train = HDF5Matrix('camelyonpatch_level_2_split_train_x.h5', 'x')
y_train = HDF5Matrix('camelyonpatch_level_2_split_train_y.h5', 'y')

datagen = ImageDataGenerator(
    preprocessing_function=lambda x: x/255.,
    width_shift_range=4, # randomly shift images horizontally
    height_shift_range=4, # randomly shift images vertically
    horizontal_flip=True, # randomly flip images
    vertical_flip=True) # randomly flip images

model.fit_generator(datagen.flow(x_train, y_train, batch_size=batch_size),
                    steps_per_epoch=len(x_train) // batch_size,
                    epochs=1024,
                    )
```

Details

Numbers

The dataset is divided into a training set of 262.144 (2^18) examples, and a validation and test set both of 32.768 (2^15) examples. There is no overlap in WSIs between the splits, and all splits have a 50/50 balance between positive and negative examples.

Labeling

A positive label indicates that the center 32x32px region of a patch contains at least one pixel of tumor tissue. Tumor tissue in the outer region of the patch does not influence the label. This outer region is provided to enable the design of fully-convolutional models that do not use any zero-padding, to ensure consistent behavior when applied to a whole-slide image. This is however not a requirement for the PCam benchmark.

Patch selection

PCam is derived from the Camelyon16 Challenge [2], which contains 400 H&E stained WSIs of sentinel lymph node sections. The slides were acquired and digitized at 2 different centers using a 40x objective (resultant pixel resolution of 0.243 microns). We undersample this at 10x to increase the field of view. We follow the train/test split from the Camelyon16 challenge [2], and further hold-out 20% of the train WSIs for the validation set. To prevent selecting background patches, slides are converted to HSV, blurred, and patches filtered out if maximum pixel saturation lies below 0.07 (which was validated to not throw out tumor data in the training set). The patch-based dataset is sampled by iteratively choosing a WSI and selecting a positive or negative patch with probability *p*. Patches are rejected following a stochastic hard-negative mining scheme with a small CNN, and *p* is adjusted to retain a balance close to 50/50.

Statistics

Coming soon

Contact

For problems and questions not fit for a github issue, please email [Bas Veeling](#).

Citing PCam

If you use PCam in a scientific publication, we would appreciate references to the following paper:

[1] B. S. Veeling, J. Linmans, J. Winkens, T. Cohen, M. Welling. "Rotation Equivariant CNNs for Digital Pathology". [arXiv:1806.03962](#)

A citation of the original Camelyon16 dataset paper is appreciated as well:

[2] Ehteshami Bejnordi et al. Diagnostic Assessment of Deep Learning Algorithms for Detection of Lymph Node Metastases in Women With Breast Cancer. JAMA: The Journal of the American Medical Association, 318(22), 2199–2210. [doi:jama.2017.14585](#)

Biblatex entry:

```
@ARTICLE{Veeling2018-qh,
  title = "Rotation Equivariant (CNNs) for Digital Pathology",
  author = "Veeling, Bastiaan S and Linmans, Jasper and Winkens, Jim and Cohen, Taco and Welling, Max",
  month = jun,
  year = 2018,
  archivePrefix = "arXiv",
  primaryClass = "cs.CV",
  eprint = "1806.03962"
}
```

Benchmark

Name	Reference	Augmentations	Acc	AUC	NLL	FROC*
GDensenet	[1]	Following Liu et al.	89.8	96.3	0.260	75.8 (64.3, 87.2)

https://github.com/basveeling/pcam

12

Name	Reference	Augmentations	Acc	AUC	NLL	FROC*
Add yours						

* Performance on Camelyon16 tumor detection task, not part of the PCam benchmark.

Contributing

Contributions with example scripts for other frameworks are welcome!

License

The data is provided under the [CC0 License](#), following the license of Camelyon16.

The rest of this repository is under the [MIT License](#).

Acknowledgements

- Babak Ehteshami Bejnordi, Geert Litjens, Jeroen van der Laak for their input on the configuration of this dataset.
- README derived from [Fashion-MNIST](#).